

Test 4: GUE KS test (full range)

Verdict: PASS | Generated: 2026-09-02 13:49:00 | Suite: AB-Cloud v19 (Julia 1.12.0)

What this test verifies

STATISTICAL family: one-sample KS of zero spacings vs GUE Wigner surmise. p-values depend on sample size and T-range ? finite-data effects are legal.

METHOD: STATISTICAL family: one-sample KS of zero spacings vs GUE Wigner surmise. p-values depend on sample size and T-range ? finite-data effects are legal.

Result

Test 4 [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

‘plots/plot_01’ (.png 600dpi / .svg / .pdf)

Plots

- ‘plots/animation.gif’ ? animated reveal of every plot of this test

Reproduce with: julia ab_cloud_v19.jl --test 4 --no-two-pass

Independent verification guide

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

[13:49:00.324] [+ 55.606s] calc: KS D = 0.021619, p_asymptotic = 9.567870e-21 (n=49999, ?=4.8368)

[13:49:00.326] [+ 55.608s] calc: KS D = 0.029562, p_asymptotic = 3.479907e-05 (n=6251, ?=2.3408)

FULL COMPUTATION LOG

[13:49:00.326] [+ 55.608s] calc: KS D = 0.025418, p_asymptotic = 6.066737e-04 (n=6250, ?=2.0125)

[13:49:00.327] [+ 55.609s] calc: KS D = 0.021305, p_asymptotic = 6.745302e-03 (n=6251, ?=1.6870)

[13:49:00.327] [+ 55.609s] calc: KS D = 0.027409, p_asymptotic = 1.619781e-04 (n=6251, ?=2.1704)

[13:49:00.328] [+ 55.610s] calc: KS D = 0.022629, p_asymptotic = 3.252259e-03 (n=6251, ?=1.7919)

[13:49:00.328] [+ 55.610s] calc: KS D = 0.019816, p_asymptotic = 1.453640e-02 (n=6251, ?=1.5691)

[13:49:00.328] [+ 55.611s] calc: KS D = 0.019647, p_asymptotic = 1.581953e-02 (n=6250, ?=1.5556)

[13:49:00.329] [+ 55.611s] calc: KS D = 0.019627, p_asymptotic = 1.595774e-02 (n=6251, ?=1.5542)

[13:49:00.331] [+ 55.613s] calc: KS D = 0.024587, p_asymptotic = 1.427535e-13 (n=24999, ?=3.8904)

[13:49:00.333] [+ 55.615s] calc: KS D = 0.018906, p_asymptotic = 3.369159e-08 (n=25000, ?=2.9916)

??

TEST 4 (secondary, HARDCORE): Objection 2: GUE Spacing Statistics ? deep KS audit

??

CONSOLE CAPTURE

Full range: N=49999 spacings, D=0.021619, p=0.000000 (report; primary verdict governs)

K1 blocks : D over 8 blocks: min=0.019627 max=0.029562, spread=0.009934 (limit 0.0101)

K2 half split: D_first=0.024587, D_second=0.018906, |?D|=0.005681 (limit 0.02)

K3 moments : mean(s)=1.0000 (limit |?|<0.02), var(s)=0.1591 vs surmise 0.1781 (limit –60%)

??

K1 block homog PASS ? spread = 0.009934

K2 half split PASS ? |?D| = 0.005681

K3 moments PASS ? mean 1.0000, var 0.1591

Test 4 [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS